

RIS-SIM v2 web dashboard — live screenshots via headless Chromium

(a) Configuration — Scenario Library cards on top, form-based editor (Form/JSON toggle per panel) below

Configuration

Room Topology

Live View

Results

Scenario Library

Click Use to load into the editor or Run to start immediately

Quickstart — Two Nodes + RIS

Minimal scenario, fast turnaround

Smallest end-to-end RIS scenario: 1 TX, 1 RX, one 8x8 RIS panel in a 10 m room at 2.4 GHz with BPSK traffic. Designed for first-time users — runs in under a second and exercises the full TX → channel → RX — impairment pipeline so the dashboard's

2.4 GHz 1 TX + 1 RX, 8x8 RIS ~1 s

RIS-SIM v2 default scenario

Use Run

Indoor Corridor (Di Renzo style)

RIS coverage extension in an office corridor

Canonical RIS-assisted indoor coverage scenario inspired by the early Di Renzo / Basar surveys: a long narrow corridor (10 × 3 × 3 m) at 2.4 GHz with

2.4 GHz 1 TX + 1 RX, 16x16 RIS, 10 m corridor ~3 s

Di Renzo et al., 2020; Basar et al., 2019 — RIS for indoor coverage

Use Run

Polar Grid Validation

Paper III-C lab reproduction

Reproduces the in-house RIS validation experiment from paper III-C: 1 TX and 30 RXs on a 45° polar grid (3 angles × 6 radii, 30 cm spacing) at 3.5 GHz, illuminating three side-by-side 5x8 RIS panels in the ip^2 (d^2 d^2) configuration with $\phi_1 = +49.64^\circ$ and

3.5 GHz 1 TX + 30 RX, 3x(5x8) RIS ~5 s

Paper III-C — in-house 5x8 RIS, E200 USRs

Use Run

mmWave Blockage Bypass

28 GHz small-cell, RIS bypassing a wall

RIS-aided mmWave hotspot from the Björnson 2020 line of work. A 28 GHz access point cannot reach a user behind a corner: a 32x32 RIS panel, mounted on the perpendicular wall, reflects the AP

28 GHz 1 AP + 1 UE, 32x32 RIS, wall bypass ~4 s

Björnson et al., 2020 — RIS in mmWave hotspots

Use Run

FM Rural Coverage

Paper Section IV - 100 MHz long-distance broadcast

Reproduces the FM rural-coverage scenario from paper Section IV: a 100 MHz transmitter at low power illuminating an 80x80 passive RIS posthorn... just past a rural village ~78 km away. The RIS reflects energy back into the village area, lifting the

100 MHz 1 TX + 5 RX, 80x80 RIS at 80 km ~4 s

Paper Section IV - passive RIS for FM broadcast extension

Use Run

Vehicular V2X (CooperRIS style)

5.9 GHz roadside RIS, mobile receiver

Vehicle-to-infrastructure scenario inspired by CooperRIS: a roadside 16x16 RIS panel (GHz BPSK) and reflects RSU signal to vehicle. The receiver uses Gauss-Markov

5.9 GHz 1 RSU + 1 vehicle (mobile), 1 ~3 s

CooperRIS-style V2X — RIS at intersection, v

Use Run

-- Topology -- -- Signal --

Start Simulation

Download JSON Upload JSON

Room

Length 10 m

Width 3 m

Height 3 m

Channel

Enable AWGN noise

Noise figure 8 dB Temp 298 K

SMALL-SCALE FADING

Enable fading

Model Rayleigh

Doppler 5 Hz K-factor 18 dB

Nodes

ID X Y Z MOBILITY SPEED (M/S)

TX 1 1.5 1.5 static 0

RX 9 1.5 1.5 static 0

+ Add node

RIS Panels

[{"id": "wall_ris", "fc": "2400000000", "type": "static", "plane": 0, "rotation": 0}]

Traffic (TX / RX Requests)

[{"mode": "transmit", "node_id": "TX", "fc": "2400000000", "sample_rate": 10000, "rotation": 0}]

Loaded scenario: indoor_corridor_2.4ghz Ticks: Avg: ETA: Samples:

(b) Results — Constellation, Channel breakdown (LOS/RIS/Total), FFT spectrum, Room heatmap

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Final Constellation

0.0004

0.0003

0.0002

0.0001

0

-0.0001

-0.0002

-0.0003

-0.0004

-0.0006

-0.0004

-0.0002

0

0.0002

0.0004

0.0006

TX

Channel Analysis (dB + dB + dB)

-10

-20

-30

-40

-50

-60

-70

LOS

RIS

Total

Boost

Power Spectrum (FFT Magnitude)

-40

-45

-50

-55

-60

-65

-70

-75

0

7

14

21

28

35

42

49

Frequency Bin

IFFT

Room Signal Heatmap (dB)

100 dB

100 100 100 100 100 100 100 100 100 100

110 dB

TX

RX

Done: 13 ticks in 0.166s Ticks: 13/14 78 t/s Avg: 2953.4 us ETA: Samples: 256

(c) Topology — interactive 3D view (orbit / pan / zoom) with wireframe room, TX/RX nodes and RIS panel

